

BIJAY THAKUR

East Elmhurst, New York

+1 (330) 8830-416

starbijay143@gmail.com

LinkedIn

GitHub

EDUCATION

Queens College, The City University of New York

Expected: Dec 2026

*Bachelor of Science in Computer Science, **GPA: 3.60***

RELEVANT COURSEWORK

- Machine Learning, Large Language Models, Data Structures & Algorithms, Linear Algebra, Probability & Statistics, Database Systems, Distributed Systems

TECHNICAL SKILLS

- Programming:** Python, SQL, JavaScript, TypeScript, Java, C++
- Machine Learning:** PyTorch, Scikit-Learn, NumPy, Pandas, Supervised Learning, Regression, Time-Series Forecasting, Anomaly Detection, Computer Vision, NLP, Feature Engineering, Train/Test Splitting, Leakage-Safe Validation, Error Analysis, MAE, RMSE, ROC-AUC, PR-AUC, Precision@K, Model Profiling
- Backend & Data:** PostgreSQL, MySQL, MongoDB, FastAPI, Flask, REST APIs, Streamlit
- DevOps & Tools:** Git, GitHub, Docker, GitHub Actions, Jupyter Notebook, VS Code, Postman

PROJECTS

GlitchVision - Visual Anomaly Detection for Game QA | GitHub

April 2026

- Built GlitchVision, a CPU-first game QA anomaly detection system using ResNet-18 embeddings, Isolation Forest, and reference-bank kNN, achieving 0.993 ROC-AUC and 0.987 PR-AUC on synthetic glitch benchmarks.
- Engineered an end-to-end ML triage pipeline with temporal scoring and Streamlit reporting, reaching 1.000 Precision@20, 1.000 interval recall, and 0.003s reference-kNN scoring latency.

Retail Demand Forecasting | GitHub

Mar 2026

- Built a forecasting pipeline using SQL + Pandas to convert invoices into weekly SKU datasets, improving ML feature quality.
- Implemented leakage-safe evaluation achieving **MAE 0.059** and **RMSE 0.594**, enabling reliable promotion-driven demand insights.

SEC 10-K RAG — Evaluated Financial QA System | GitHub

May 2026

- Built a citation-grounded SEC 10-K RAG pipeline with structured HTML/XBRL parsing, OpenAI embeddings, ChromaDB, BM25, hybrid RRF retrieval, and cross-encoder reranking over Apple's 2025 annual report.
- Implemented a parallel LlamaIndex version to compare framework vs. from-scratch RAG design, reducing core pipeline code by 22% while achieving 0.74 Recall@1, 0.8032 MRR, and 0.9901 RAGAS faithfulness on a 50-question gold eval set.

EXPERIENCE

AI Agent Builder Intern, NeuralSeek

April 2026 - Present

Remote

- Built and refined AI agent workflows on NeuralSeek's platform, focusing on multi-step reasoning, automation, and output validation.
- Evaluated agent behavior across edge cases, identified failure points, and improved workflow reliability through testing and iteration.

Academic Tutor, LaGuardia Community College

Feb 2024 - Jun 2024

Long Island, NY

- Mentored students 1-on-1 in C++ and algorithmic problem-solving, resulting in improved code debugging skills.
- Created structured notes and practice modules that helped students reduce coding errors and understand core CS concepts faster.

Web Developer Intern, WebX Nepal

Mar 2022 - Jul 2022

Kathmandu, Nepal

- Developed responsive web pages using HTML, CSS, and JavaScript, enhancing website usability and cross-device performance.
- Collaborated with developers to transform client requirements into functional UI components and improve UX.